

ANAÏS AZOUAOU

MSc student in AI & Robotics, seeking a 6-month research internship from March 2027

Research interests: Quality-Diversity, evolutionary robotics, LLM-guided evolutionary search, robot learning

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EDUCATION

Sorbonne University

Paris

Master's in Computer Science, Algorithms, AI, Interactions & Decision (AI2D)

2025 – 2027 (expected)

Sorbonne University

Paris

BSc in Computer Science

2022 – 2025

RESEARCH EXPERIENCE

Interpretable ML for Intensive Care | *Research Internship*

Jun – Aug 2026

IRIT, REVA team - CHU de Toulouse, Toulouse

Supervisors: S. Cussat-Blanc, D. Wilson, G. Nadizar

- Modeled clinician-rated severity of 4 respiratory disorders in 164 ventilated ICU patients.
- Benchmarked 16 ML models (linear, kernel, trees, MLP), reaching 82–96% of the inter-rater ceiling.
- Built a **Cartesian GP** engine (with **NSGA-II** variant), benchmarked against PySR, gplearn and hal-cgp.
- Matched **Extra Trees** (best ML model) with 2–3 variable formulas: 94–111% of its R^2 across disorders.
- Tested 5 **LLM-assisted formula searches**, the best beat Extra Trees on a disorder (R^2 0.86 vs. 0.81).

Real-to-Sim Scene Generation | *Research Project*

Feb – May 2026

ISIR, Sorbonne University, Paris

Supervisors: S. Doncieux, E. Garrabe

- Built a pipeline turning a text prompt or a photo into a physically valid 3D scene for **Genesis**.
- Fine-tuned **Phi-3-mini** with **QLoRA** on 3,531 synthetic examples to run fully locally.
- Improved spatial-relation F1 from 56% (GPT-4o pipeline) to 72% at equal latency.
- Reduced position error from 1.25 to 0.48 m on real Franka Panda scenes with **GroundingDINO**.

PROJECTS

Damage Recovery for a Quadruped Robot

- Built Ant gait repertoires with **MAP-Elites**, **PGA-ME** and **DCRL-ME** (JAX, QDax), 512k episodes each.
- Evaluated repertoires: DCRL-ME widest (84% coverage), PGA-ME best gaits left after damage.
- Adapted to 12 damage scenarios with **Intelligent Trial & Error** (ITE) in a median of 5 trials.
- Benchmarked **TD3**: fine-tuning matched ITE only with a critic warm-up (8/8 runs, ~450 episodes); a damage-robust policy matched it with zero trials.
- Found that trying gaits by intact performance nearly matches ITE: repertoire quality matters most.

BirdCLEF+ 2026 Audio Classification

- Developed a **multi-label classification** pipeline for 206 species from noisy, highly imbalanced audio.
- Reached 0.95 ROC-AUC with frozen **Perch embeddings**, vs. 0.84 with 969 **handcrafted features**.
- On real field soundscapes, raised the handcrafted model from 0.58 to 0.70 with **pseudo-labelling**.

Sign Language Recognition

- Trained a **Conv-Transformer** on **MediaPipe** landmarks to recognize 250 ASL and 100 LSFB signs.
- Reached 64.8% (ASL) and 68.3% (LSFB) top-1 on unseen signers, 20 points above a **linear baseline**.
- Ran 10 **one-factor ablations**, identifying dominant-hand canonicalization as the key step (+3.8 points).

SKILLS

Evolution: Evolutionary Robotics, Quality-Diversity, Genetic Programming, Symbolic Regression

Machine Learning: Reinforcement Learning, Robot Learning, Deep Learning, LLMs

Mathematics: Multi-objective & Bayesian Optimization, Probabilistic Models & Statistics, Linear Algebra

Tools: Python, JAX (QDax, Brax), DEAP, PyTorch, Hugging Face, scikit-learn, Genesis, C++, Git

Languages: French (native), English (proficient), Arabic (proficient)

PUBLICATIONS

- A. Azouaoui**, C. Zouloumian, E. Garrabe, S. Doncieux. *Real-to-Sim: generating physically valid robot simulation scenes from text and images with LLMs and VLMs*. In preparation.
- A. Azouaoui**, G. Nadizar, S. Cussat-Blanc, D. Wilson, M. Poette. *Interpretable prediction of clinician-rated respiratory severity in intensive care with genetic programming*. In preparation.